A BOUNDED BELOW OBJECT PULLED BACK FROM ITS PUSHFORWARD BY A FLAT MORPHISM OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete reduction: the passage to cohomology sheaves, the flatness computation and the membership in the subcategory D' are all in the counted span.

Editorial note. The label lemma-bounded-in-image-upstairs occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Flat morphism of ringed topoi and exactness of its pullback. Same chapter, the definition of a flat morphism of ringed topoi. Supplies the hypothesis under which f^* commutes with the passage to cohomology sheaves used in the counted proof.
- Cohomology sheaves and truncations tau_{<= j} of an object of D(O_C). Same chapter, the truncation functors and the fact that an object is zero if and only if all its cohomology sheaves vanish. Supplies the reduction to the bounded case and the step H^j(Rf_* K) = H^j(Rf_* tau_{<= j} K).
- The triangulated subcategory D' of objects whose cohomology sheaves satisfy f^*Rf_*H^q = H^q. Same chapter, the subcategory D' introduced in the upstairs lemma and its stability under distinguished triangles. Supplies the final step, in which K is placed in D' because all its cohomology sheaves are.

Source reference. Original lemma 0D7R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6315–6315. Same-chapter navigation pointer invoked at proof line 6364, where the author places the cohomology sheaves in the triangulated subcategory introduced by that earlier lemma. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Let K be an object of $D(\mathcal{O}_\mathcal{C})$. Assume*

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(1) *f is flat,*
(2) *K is bounded below,*
(3) *$f^*Rf_*H^q(K) \to H^q(K)$ is an isomorphism.*

Then $f^*Rf_*K \to K$ is an isomorphism.

Proof. Observe that $f^*Rf_*K \to K$ is an isomorphism if and only if it is an isomorphism on cohomology sheaves H^j. Observe that $H^j(f^*Rf_*K) = f^*H^j(Rf_*K) = f^*H^j(Rf_*\tau_{\leq j}K) = H^j(f^*Rf_*\tau_{\leq j}K)$. Hence we may assume that K is bounded. Then property (3) tells us the cohomology sheaves are in the triangulated subcategory $D' \subset D(\mathcal{O}_\mathcal{C})$ of Lemma 0D7R. Hence K is in it too. □